VISTRA

COMANCHE PEAK NUCLEAR POWER PLANT

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CP-202400097
TXX-24020
May 6, 2024

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Ref 10CFR50.73

Subject: Comanche Peak Nuclear Power Plant (CPNPP)
Docket Nos. 50-445 and 50-446
Potential Loss of Safety Function Due to the Inoperability of Both Trains of Uninterruptible Power Supply Heating, Ventilation, and Air Conditioning (UPS HVAC)
Licensee Event Report 1-24-001-00

Dear Sir or Madam:

Attached is Licensee Event Report (LER) 1-24-001-00, " Potential Loss of Safety Function Due to the Inoperability of Both Trains of Uninterruptible Power Supply Heating, Ventilation, and Air Conditioning (UPS HVAC)" for Comanche Peak Nuclear Power Plant (CPNPP) Units 1 and 2.

This Communication contains no new commitments regarding CPNPP Units 1 or 2.

Should you have any questions, please contact Ryan Sexton at (979) 292-5064 or ryan.sexton@vistracorp.com.

Sincerely,

Steven K. Sewell

Attachment: NRC Form 366

c(email) - John Monninger, Region IV [John.Monninger@nrc.gov]
Samson Lee, NRR [Samson.Lee@nrc.gov]
John Ellegood, Senior Resident Inspector, CPNPP [John.Ellegood@nrc.gov]
Henry Strittmatter, Resident Inspector, CPNPP [Henry.Strittmatter@nrc.gov]

NRC FORM 366 (04-02-2024) U.S. NUCLEAR REGULATORY COMMISSION

APPROVED BY OMB: NO. 3150-0104 EXPIRES: 04/30/2027

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name		2. Docket Number	3. Page
Comanche Peak Nuclear Power Plant	☒ 050 / ☐ 052	445	1 OF 6

4. Title
Potential Loss of Safety Function Due to the Inoperability of Both Trains of the Uninterruptible Power Supply Heating, Ventilation, and Air Conditioning (UPS HVAC)

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name		Docket Number
03	05	2024	2024	- 001 -	00	05	06	2024	CPNPP	☒ 050	446
									Facility Name	☐ 052	Docket Number

9. Operating Mode	10. Power Level
Mode 1	100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20		10 CFR Part 50			
☐ 20.2201(b)	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(d)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2203(a)(1)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(2)(i)	☐ 20.2203(a)(4)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(ii)	**10 CFR Part 21**	☐ 50.69(g)	☐ 50.73(a)(2)(v)(A)	**10 CFR Part 73**	☐ 73.1200(e)
☐ 20.2203(a)(2)(iii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☑ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
			☐ 50.73(a)(2)(vii)		

☐ **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Ryan Sexton	979-292-5064

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
X	EF	MO	W120	Y	D	EF	FCU	E&W	N

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☑ No	☐ Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)
On March 05, 2024, Units 1 and 2 were operating at 100% power. At 0720 the Uninterrupted Power Supply, Heating, Ventilation, and Air Conditioning (UPS HVAC) X-02 automatically shutdown, which caused the station to determine that there was a potential loss of safety function for the equipment in the UPS and Distribution Rooms, due to losing the ability to control the temperature during a postulated emergency condition. This meets the safety system functional failure criteria. This was due to the failure of the UPS HVAC X-02 unit, concurrent with the UPS HVAC X-01 unit and the Emergency Fan Coil Units (EFCUs) being out of service for maintenance. The immediate corrective actions were for the station to start the EFCUs at 0729 which restored cooling to three of the four UPS and Distribution Rooms. This restored the safety function after approximately nine minutes. The station then restored operability to the UPS HVAC X-01 system by switching the cooling system to the other unit (Unit 2, Train A) approximately 41 minutes after the UPS HVAC X-02 trip, which restored the temperature control to all four UPS and Distribution Rooms.
The root cause of this event was the station exhibiting a less than adequate conservative bias to the unique operability concerns associated with the UPS HVAC Cooling System when performing off-normal maintenance.

NRC FORM 366B (04-02-2024)

U.S. NUCLEAR REGULATORY COMMISSION

LICENSEE EVENT REPORT (LER)

(FAILURE CONTINUATION)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Comanche Peak Nuclear Power Plant	☑ 050 ☐ 052	445	24	001	00

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANUFACTURER	REPORTABLE TO IRIS
D	EF	COND	GMC	N

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER		
			YEAR	SEQUENTIAL NUMBER	REV NO.
Comanche Peak Nuclear Power Plant	☒ 050 ☐ 052	445	24	- 001 -	00

NARRATIVE

I. DESCRIPTION OF REPORTABLE EVENT

On March 05, 2024, Units 1 and 2 were operating at 100% power. At 0720 the Uninterrupted Power Supply, Heating, Ventilation, and Air Conditioning (UPS HVAC) X-02 unit tripped, which caused the station to potentially lose a safety function. Specifically, the station lost the ability to properly control the temperature of the UPS and Distribution Rooms during a postulated emergency condition. The other factors leading to this were the Emergency Fan Coil Units (EFCUs) being off, which renders them inoperable, and UPS HVAC X-01 being inoperable, due to the support system, Component Cooling Water (CCW) Unit 1, Train A (1-A), being inoperable . The station restored the safety function after approximately nine minutes, by starting all of the EFCUs at 0729, which restored cooling to three of the four UPS and Distribution Rooms (EFCU 1-A was only functional, due to the support system CCW 1-A being inoperable, however forced air circulation was returned). The station then connected UPS HVAC X-01 to CCW Train 2-A approximately 41 minutes after the event, which restored temperature control to all four UPS and Distribution Rooms.

A. REPORTABLE EVENT CLASSIFICATION

This event is reportable under 10CFR50.73(a)(2)(v)(D), "Any event or condition that could have prevented the fulfillment of the safety function of the structures or systems that are needed to ... Mitigate the consequences of an accident." This is based on the station losing both trains that provide temperature control to the UPS and Distribution Rooms, during a postulated Large-Break Loss of Cooling Accident (LBLOCA), or a similar postulated event which causes a high containment pressure, HI-3, signal (18.0 psig). This is due to losing all three cooling systems in the UPS and Distribution Rooms, specifically:

1) The cooling system supporting UPS HVAC X-01, CCW Train 1-A, being inoperable for planned maintenance on the heat exchanger (Hx). Even-though CCW is cross-connected to the 1-B train for all alternative conditions, a HI-3 signal would cause CCW to isolate Train 1-A from Train 1-B to support containment spray.
2) The four EFCUs being turned off at the time of the event, based on procedural allowance, which renders them inoperable due to no auto-start mechanism.
3) UPS HVAC X-02 being inoperable due to a failure of the evaporator blower motor, causing the system to shutdown.

B. PLANT CONDITION PRIOR TO EVENT

On March 05, 2024, prior to the event, both Units 1 and 2 were operating at approximately 100% power.

C. STATUS OF STRUCTURES, SYSTEMS, OR COMPONENTS THAT WERE INOPERABLE AT THE START OF THE EVENT AND CONTRIBUTED TO THE EVENT

The event caused the station to immediately lose UPS HVAC X-02, due to the motor shutdown The EFCUs in the UPS and Distribution Rooms were inoperable prior to the event, due to being turned off, based on maintenance personnel working in UPS and Distribution Room associated with Train 1-A. UPS HVAC X-01 was inoperable prior to the event, based on the CCW Train 1-A Hx maintenance.

D. NARRATIVE SUMMARY OF THE EVENT, INCLUDING DAYS AND APPROXIMATE TIMES

On March 05, 2024, Units 1 and 2 were operating at 100% power. At 0720 a failure occurred for UPS HVAC X-02 [EIIS: (EF)(ACU)], specifically, the evaporator blower motor [EIIS: (EF)(MO)] had a immediate failure. This event, coupled with the EFCUs [EIIS: (EF)(FCU)] being inoperable and the support system for UPS HVAC X-01 being inoperable, CCW Train 1-A Hx [EIIS (CW)(Hx)], resulted in the station losing all trains of UPS HVAC, and hence potentially losing the safety function to mitigate consequences of a postulated accident by losing the ability to control the temperature of the UPS and Distribution Rooms during an accident.

NARRATIVE

The UPS HVAC system is comprised of two trains, each of which is composed of one Air-Conditioner (A/C), with an auto-start feature, which can provide 100% cooling capacity in all four UPS and Distribution Rooms, and two EFCUs (one per UPS and Distribution Room) each of which can provide 100% cooling to the associated UPS and Distribution Room, when powered on. The UPS HVAC Train is operable when either the associated A/C is operable or the two train-specific EFCUs are operable (i.e., UPS HVAC X-01 or EFCU Train 1-A and Train 2-A).

Operators responded to the event immediately to return the safety function of the UPS HVAC system by turning on the EFCUs. Both Units 1 and 2 continued to operate at 100% power in MODE 1 for the entirety of the event. The UPS and Distribution Rooms did not have any measurable temperature change, so there was no challenge of the UPS HVAC auto start function.

E. THE METHOD OF DISCOVERY OF EACH COMPONENT OR SYSTEM FAILURE, OR PROCEDURAL PERSONNEL ERROR

The control room operators received common window X-ALB-11C UPS & DISTR RM A/C SYS FN TRN A/B DELTA-P LO alarm, which notified the operators of the loss of UPS HVAC X-02. The control room operators were also aware of both the inoperability of UPS HVAC Train 1-A, due to the inoperability of the support system CCW Train 1-A, and the EFCUs being turned off for the maintenance work, which made them inoperable. Turning off the EFCUs when maintenance is located in the UPS and Distribution Rooms is procedurally allowed, as long as one of the two UPS HVAC units is running. Losing UPS HVAC X-02, and the concurrent loss of all redundant cooling in the UPS and Distribution Rooms caused the station to potentially lose the safety function of equipment in the UPS and Distribution Rooms, due to a loss of temperature control.

II. COMPONENT OR SYSTEM FAILURES

A. CAUSE OF EACH COMPONENT OR SYSTEM FAILURE

The UPS HVAC X-01 inoperability was caused by the station performing planned maintenance on the CCW Train 1-A Hx, and the station not realigning UPS HVAC X-01 to Train 2-A.

The UPS HVAC X-02 failure was caused by the immediate failure of the evaporator blower motor, while the system was in service.

The EFCU failure was caused by planned maintenance which turned off all EFCUs to assist the maintenance team working in the UPS and Distribution room.

B. FAILURE MODE, MECHANISM , AND EFFECTS OF EACH FAILED COMPONENT

The control room staff was previously aware of both the fact that UPS HVAC X-01 was inoperable due to the support system not being connected to CCW Train 2-A, as well as that the decision was made to turn off all EFCUs. The control room operators were made aware of the immediate UPS HVAC X-02 evaporator blower motor failure due to alarms in the control room, which led them to immediately send control room operators down to the UPS and Distribution Rooms to locally turn all four of the EFCUs back on, approximately nine minutes later. The effects of all trains of UPS HVAC being declared inoperable concurrently was that the station lost the ability to appropriately control the temperature in the UPS and Distribution rooms, and the potential loss of the safety functions performed by equipment in this room. There was no measurable temperature change occurring from the event.

C SYSTEMS OR SECONDARY FUNCTIONS THAT WERE AFFECTED BY FAILURE OF COMPONENTS WITH MULTIPLE FUNCTIONS

The UPS HVAC system does not have a secondary function beyond providing temperature control for the safety related UPS and Distribution Rooms during all normal and accident conditions.

NARRATIVE

D. FAILED COMPONENT INFORMATION

The UPS HVAC X-02 blower motor is a Westinghouse, open drip proof, squirrel cage induction motor. The EFCUs are Ellis and Watts, model number ACH-100, with serial numbers 14500 to 145503. The UPS HVAC X-01 failed due to the inoperability of the support system CCW Train which provides cooling to the condenser of the UPS HVAC. The condenser is a Graham Manufacturing Company, tube and shell, single pass component.

II. ANALYSIS OF THE EVENT

A. SAFETY SYSTEM RESPONSES THAT OCCURRED

The UPS HVAC X-01 system was functional through the entire event due to the cross connection of CCW to Train 1-B, however the system did not start, due to no measurable temperature rise through the event. The operators started all of the EFCUs in the UPS and Distribution Rooms in approximately nine minutes, which restored the temperature control function to three of the four UPS and Distribution rooms. The operators then connected the UPS HVAC X-01 to CCW Train 2-B approximately 41 minutes post-event, which restored the temperature control function to all UPS and Distribution rooms.

B. DURATION OF SAFETY SYSTEM TRAIN INOPERABILITY

The station lost both trains of UPS HVAC for approximately nine minutes, 0720 to 0729, which lost the stations ability to control the temperature in the UPS and Distribution Rooms in all conditions. At 0729 the station restored the operability of UPS HVAC Train B by restoring all Train B EFCUs to operable, however UPS HVAC Train A was still inoperable due to EFCU Train 1-A and UPS HVAC Train 1-A both having support system inoperability. At 0801, the station connected UPS HVAC X-01 to CCW Train 2-A to restore operability to Train A of the UPS HVAC system.

C . SAFETY CONSEQUENCES AND IMPLICATIONS OF THE EVENT

In this event, the station continued to operate both Units at 100% power and stayed within the technical specification 3.7.20 limits for having the UPS HVAC trains inoperable. Therefore, this event was an American Nuclear Society Condition I occurrence (Normal Operation and Operation Transients) as a deviation which may occur during continued operation as permitted by the plant Technical Specifications (FSAR Section 15).

The UPS HVAC is capable of removing sensible and latent heat loads from the UPS inverter rooms, which include consideration of equipment heat load requirements to ensure equipment operability per the Technical Specification Bases.

The station lost the safety function to mitigate the consequences of a postulated LBLOCA, or a similar event which causes a HI-3 pressure signal. This is an American Nuclear Society Condition IV occurrence (Limiting Fault). This potential is increased for this condition, as the operations staff has no procedure-driven actions to restore the EFCUs to operation or connect the X-01 UPS HVAC to Unit 2 during this occurrence.

The operators started all of the UPS and Distribution Room EFCUs approximately nine minutes into the event which restored the temperature control function of the UPS HVAC system to three of the rooms, barring the one associated with Unit 1 Train A due to the EFCU 1-A still being inoperable. The station then connected the UPS HVAC X-01 to CCW Train 2-A approximately 41 minutes after the UPS HVAC X-02 trip, restoring the temperature control function to all UPS HVAC rooms.

Since Unit 1 CCW trains were cross-connected, and no event occurred to cause the CCW trains to separate (i.e., a LBLOCA with Containment Spray Actuation), UPS HVAC train X-01 was functional for temperature control for the entire event. This event is not considered to have had any significant effect on the health and safety of the public.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Comanche Peak Nuclear Power Plant	■ 050 / ☐ 052	445	24	001	00

NARRATIVE

IV. CAUSE OF THE EVENT

The root cause of this event was station personnel exhibiting a less than adequate conservative bias to the unique operability concerns associated with the UPS Cooling System when performing off-normal maintenance (i.e., at-power work requiring the inoperability of CCW Train 1-A).

The causal analysis is still ongoing, and if substantial information is identified that would significantly change the understanding of the course or consequences of the event, or if there is a substantial change to the root cause or the corrective actions planned by the station, then a supplemental report will be sent.

V. CORRECTIVE ACTIONS

The immediate corrective actions were to turn on all four EFCUs and connect UPS HVAC X-01 to CCW Train 2-A to restore operability to both trains of UPS HVAC. The station determined that this was a significant condition adverse to quality and performed a Root Cause Evaluation on the Event.

The planned future Corrective Actions include two Corrective Action to Prevent Recurrence (CAPRs):

1) Revise the applicable procedure(s) to ensure that when a system that provides support to unit-common equipment is inoperable then impact to the opposite unit should be evaluated per the Safety Function Determination Program (SFDP) for potential cross-unit interface, and that if unit-common safety function is impacted, the appropriate LCO actions should be taken for both units (i.e., the SFDP is not applicable).

2) Revise the applicable procedure(s) to ensure that for systems that support common loads (CCW, SSW, CHS, etc.), a list of common loads supplied by the system to be taken out of service should be listed in the clearance notepad.

The other future corrective actions include performing a case study of this event to be presented to all relevant departments.

VI. PREVIOUS SIMILAR EVENTS

On August 20, 2003 Comanche Peak was forced to declare both trains of the Control Room Air Conditioning (CRAC) system inoperable. This is a common system that is fed by safeguard CCW cooling. CRAC is comprised of two trains, each with two 50% capacity Air-Conditioning Units. This event can be referenced as LER 445/03-004-00 (ML033000104). The root cause of this event was not determined to be related to the common CCW system being appropriately aligned. Based on this decision the CAPRs that came out of this RCE would not be expected to have prevented this event.